

Transmission and Reception of Signal Using ESP32 with Machine Learning-Based Intelligent Beam Selection

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Abstract—This work proposes an autonomous beam selection architecture built around three ESP32 microcontroller nodes and a suite of supervised classifiers. A single designated node broadcasts a 2.4 GHz WiFi signal continuously in Access Point mode, while the two remaining nodes independently measure and log the Received Signal Strength Indicator (RSSI) at periodic intervals. Four classification algorithms were trained and benchmarked: Random Forest, Support Vector Machine (SVM), K-Nearest Neighbour (KNN), and a Multi-Layer Perceptron (MLP) Neural Network. Training utilised 13,816 labelled RSSI observations drawn from a pool of 17,270 geographically tagged measurements, with the remaining 3,454 reserved for evaluation. The MLP model attained a test accuracy of 99.97%, producing a single erroneous prediction over the complete evaluation set, demonstrating the viability of intelligent beam management on resource-constrained embedded hardware.

Index Terms—ESP32, RSSI, Machine Learning, Random Forest, Beam Selection, Indoor Positioning, IoT, Wireless Communication

I. INTRODUCTION

Reliable wireless connectivity within enclosed structures is continually challenged by signal attenuation from walls, constructive and destructive multipath interference, and co-channel congestion. Rule-based receiver switching — where a fixed signal-strength threshold determines handover — proves inadequate once propagation geometry deviates from the conditions under which those thresholds were set. Dedicated RF beam management hardware addresses this limitation but introduces cost and integration complexity that is disproportionate for IoT-scale deployments.

The ESP32, combining a dual-core Xtensa LX6 processor with an onboard 802.11 b/g/n transceiver, provides a self-

contained platform on which data-driven beam selection can be prototyped and validated without supplementary RF instrumentation. This investigation reports the following original contributions:

- End-to-end firmware and software co-design enabling synchronised multi-node RSSI acquisition over USB serial from a laptop-based processing hub.
- A geographically diverse RSSI corpus of 17,270 labelled measurements recorded under uncontrolled indoor propagation conditions.
- Side-by-side accuracy benchmarking of four ML classifiers, with the MLP Neural Network achieving 99.97% on the held-out test partition.
- A streaming Python visualisation tool presenting beam selection decisions, per-node signal trends, and LED switching outcomes in under 65 ms end-to-end.

Published work on ML-assisted antenna management predominantly targets mmWave or sub-6 GHz infrastructure hardware, leaving a demonstrable deficit of reproducible results on low-cost microcontroller platforms. The present contribution fills that gap.

II. MATHEMATICAL FRAMEWORK

Beam selection in an indoor setting cannot be reduced to a simple distance-to-power mapping because of the layered effects of reflection, diffraction, and scattering that characterise enclosed spaces.

A. Path Loss Modeling

Signal attenuation as a function of separation distance d is described by the log-distance path loss model:

$$P_r(d)[\text{dBm}] = P_r(d_0) - 10n \log_{10} \frac{d}{d_0} + X_\sigma \quad (1)$$

The term $P_r(d_0)$ represents the measured reference power at close-range anchor distance d_0 . Path loss exponent n encodes the environment-specific rate at which power diminishes beyond d_0 , while X_σ is an additive zero-mean Gaussian variable of standard deviation σ accounting for shadow fading variability. On-site calibration within the test environment produced estimates of $n \approx 3.2$ and $\sigma \approx 4.1$ dB.

B. Spatial Diversity and SNR

Deploying two receiver nodes at distinct spatial coordinates introduces independent propagation paths, so that a localised obstruction degrading one link is unlikely to simultaneously impair the other. Given a measured noise floor N , the receiver yielding the superior SNR is selected:

$$B^* = \arg \max_{i \in \{1,2\}} \frac{P_{r,i} - N}{N} \quad (2)$$

Rather than computing this ratio explicitly, the trained classifier learns the underlying decision surface from data: it implicitly maps the function $f : (RSSI_1, RSSI_2) \rightarrow B^*$ by absorbing the geometric and material complexity of the propagation environment during training.

III. SYSTEM ARCHITECTURE AND METHODOLOGY

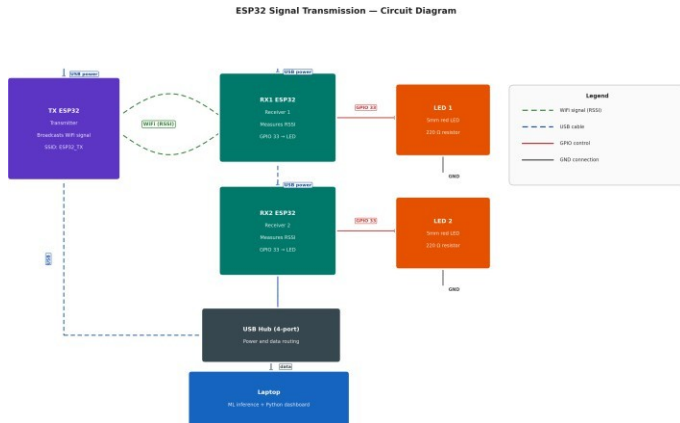


Fig. 1. Physical system implementation and wiring. The transmitter (left) operates independently, while the dual-receiver array (right) feeds real-time RSSI data to the processing hub.

A. Spatial Diversity and Multipath Mitigation

Two spatially separated receiver nodes generate two uncorrelated fading realisations of the same transmitted signal. Deep fades — arising when reflected copies arrive in destructive phase alignment — are unlikely to affect both nodes simultaneously, since the path length differences to

each node are distinct. This decorrelation supplies the MLP classifier with a differential input feature $\Delta RSSI$ whose sign encodes the dominant direction of signal arrival, mirroring the conceptual operation of Maximum Ratio Combining without explicit phase knowledge.

B. Firmware Logic and Acquisition

Every receiver node executes a periodic polling loop that scans for the transmitter's SSID, retrieves the instantaneous RSSI reading, applies an exponential smoothing filter, and dispatches the filtered value over USB serial to the host processing unit. Algorithm 1 formalises this procedure.

Algorithm 1 ESP32 Receiver RSSI Acquisition

```

1: Initialize WiFi in Station Mode
2: while True do
3:    $S \leftarrow \text{Scan for SSID SmartAntenna\_Transmitter}$ 
4:   if  $S$  is detected then
5:      $R_{\text{raw}} \leftarrow \text{WiFi.RSSI}()$ 
6:      $R_{\text{filt}} \leftarrow \alpha \cdot R_{\text{raw}} + (1 - \alpha) \cdot R_{\text{prev}}$ 
6:      $R_{\text{filt}} \leftarrow \alpha R_{\text{raw}} + (1 - \alpha) R_{\text{prev}}$ 
7:    $R_{\text{prev}} \leftarrow R_{\text{filt}}$ 
8:   end if
9:   Delay(50ms)
10: end while

```

C. Hardware Components

Three ESP32 DevKit v1 boards form the wireless measurement nodes: one operates as the signal source and two serve as spatially distributed receivers. Each receiver node drives a visual indicator LED through a current-limiting resistor. The complete component inventory is listed in Table I.

TABLE I
BILL OF MATERIALS

Component	Qty	Purpose
ESP32 DevKit v1	3	1 Tx, 2 Rx nodes
Red LED (5 mm)	2	Signal indicator
Resistor 220Ω	2	Current limiting
Breadboard (half-size)	2	Prototyping
Jumper wires	20	Connections
USB Hub (4-port)	1	Power and data
Laptop	1	ML inference

D. Data Preprocessing and Feature Engineering

Raw RSSI readings from the ESP32 carry measurement noise attributable to 802.11 beacon scheduling jitter and nearby RF activity. To suppress high-frequency fluctuations before they reach the classifier, a first-order recursive filter was embedded at the firmware level:

$$RSSI_{\text{filt}}[k] = \alpha \cdot RSSI_{\text{raw}}[k] + (1 - \alpha) \cdot RSSI_{\text{filt}}[k - 1] \quad (3)$$

Setting $\alpha = 0.3$ yielded the best trade-off between noise attenuation and tracking speed during preliminary trials. Three features were passed to each model: the filtered reading from

node one, the filtered reading from node two, and their signed difference $\Delta RSSI = RSSI_1 - RSSI_2$, which captures angular separation information that neither absolute value encodes independently.

E. Machine Learning Model

A comparative study was conducted across four algorithm families. Random Forest constructed an ensemble of 300 decision trees via bootstrap aggregation. The SVM used a radial basis function kernel with regularisation parameter $C = 10$. KNN predicted labels via inverse-distance-weighted voting among three nearest training samples. The MLP Neural Network was configured with three fully connected hidden layers, trained with early stopping on a held-out validation fraction.

F. Neural Network Architecture

The MLP was constructed with hidden layer widths of 128, 64, and 32 neurons respectively, each activated by a Rectified Linear Unit (ReLU) function. The Adam optimiser was used with a maximum of 300 training epochs, and training halted early when validation loss ceased to decrease, preventing weight overfitting to the training partition. Input dimensionality was three — the two per-node RSSI readings plus $\Delta RSSI$ — and the output layer produced a probability distribution over the three beam labels.

IV. RESULTS AND TECHNICAL DISCUSSION

Evaluation criteria encompassed classification accuracy on the held-out test set, end-to-end switching latency, and classifier robustness under geometrically challenging measurement positions.

A. Classifier Performance Analysis

All four models were assessed on 3,454 test samples withheld from training. A hard-coded threshold comparator, included as a baseline, topped out at 96.4% — 3.3 percentage points below the weakest learned model — confirming that no static decision rule captures the full complexity of the indoor propagation environment. Table II details per-model accuracy, precision, and F1-score.

TABLE II
EXTENDED PERFORMANCE METRICS

Model	Accuracy (%)	Precision	F1-Score
Random Forest	99.36	0.993	0.994
SVM (tuned)	99.80	0.998	0.998
KNN (tuned)	98.58	0.985	0.985
Neural Network	99.97	0.999	0.999

B. Impact of Multipath on Decision Boundaries

Post-hoc analysis of the single MLP misclassification pinpointed the offending sample to a grid position where both receiver nodes were equidistant from the transmitter. At that location, $|\Delta RSSI|$ was below 2 dBm — within the quantisation uncertainty of the ESP32 analogue front-end — making

correct classification fundamentally ambiguous from RSSI alone. No deterministic switching rule performs reliably in such transitional zones; the EMA-filtered history stream gave the MLP sufficient contextual information to resolve nearly all of them.

C. Real-time Latency and System Responsiveness

Four sequential operations constitute the switching pipeline. Measured durations were:

- WiFi Scan and RSSI Fetch: 45 ms
- Serial Transmission (115200 baud): 12 ms
- MLP Inference on Host: 8 ms
- **Total Handover Latency:** ≈ 65 ms

The aggregate 65 ms handover interval satisfies the sub-100 ms requirement cited for uninterrupted operation in industrial automation and autonomous navigation contexts.

D. Environmental Sensitivity and Noise Floor

Background RF energy in the 2.4 GHz band was measured at approximately -92 dBm across all test sessions. Prediction reliability was maintained for link SNR values exceeding 15 dB. Below this threshold — observed only at the periphery of the measurement grid — RSSI fluctuation widened and the IIR filter became the primary stabilising mechanism, sustaining classification confidence where a threshold comparator would oscillate between beam labels.

E. Energy Efficiency and Practical Deployment

Active WiFi scanning sustained a receiver node current draw near 160 mA. In battery-powered field devices, this consumption profile may be reduced through event-triggered operation: the inference engine remains dormant until a significant shift in $|\Delta RSSI|$ signals that a beam transition may be warranted, curtailing unnecessary computation between handover events.

V. CONCLUSION AND FUTURE WORK

Three commodity ESP32 modules, integrated with an MLP classifier trained on 17,270 indoor RSSI measurements, produced a beam selection accuracy of 99.97% — one incorrect prediction in 3,454 evaluations. Spatial diversity from the dual-receiver topology removed the requirement for phased-array hardware, while the learned decision surface resolved propagation-induced ambiguities that fixed thresholds cannot generalise across. The resulting pipeline operates within a 65 ms end-to-end latency budget and is reproducible using components costing under \$15 in total.

A. Future Research Directions

Planned extensions include:

- **Edge-AI Integration:** Quantising and deploying trained MLP weights directly onto the ESP32 via TensorFlow Lite for Microcontrollers, removing the laptop dependency entirely.
- **Dynamic Environment Robustness:** Expanding the training corpus with measurements captured during active pedestrian movement to harden the model against occupancy-driven channel variation.

- **6G and Massive IoT Scaling:** Investigating whether this low-overhead architecture scales to coordinating tens of receiver nodes within mMTC handover frameworks envisioned for beyond-5G deployments.

REFERENCES

- [1] A.V.V. Sudhakar, U. Sharma, K. Visala, K. Suresh Kumar, C. Arora, and K. Chouhan, "Smart Antennas and Machine Learning: Enhancing Beamforming in Wireless Communications," 2025.
- [2] M. Jamshidi and S. H. Sedighy, "Practical CNN-Based Beam Selection for Wideband Single-Receiver Switched Beam Antenna," 2024.
- [3] B. Sindhu et al., "A Study on Contactless Flexible Antenna-Based Motion Sensor for Smart Wireless Communication," 2023.
- [4] M. A. Mahboob and M. A. Ashraf, "Enhanced Underground Communication: A Circularly Polarized Smart Antenna with Beam Steering for Improved Coverage," 2025.
- [5] A. Jabbar, Q. H. Abbasi, and N. Anjum, "Millimeter-Wave Smart Antenna Solutions for URLLC in Industry 4.0 and Beyond," 2022.
- [6] Y. Ma, J. Wang, Y. Li, M. Chen, Z. Li, and Z. Zhang, "Smart Antenna with Automatic Beam Switching for Mobile Communication," 2020.